VHDL 과제2

로봇공학과 2023079925 최소영

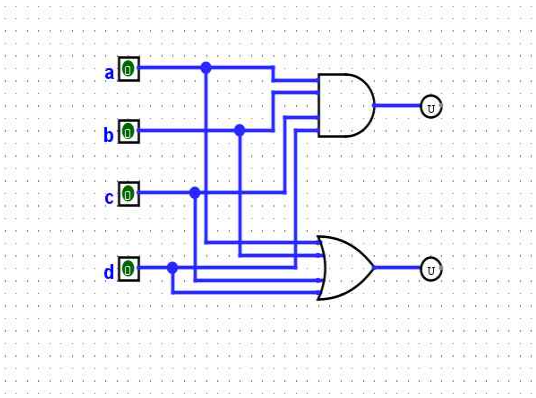
1. Write VHDL code.

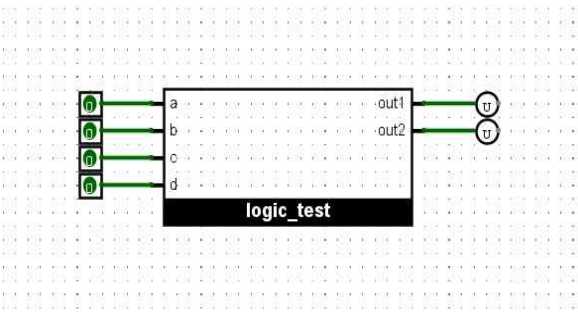
```
--library and package
library ieee;
use ieee.std_logic_1164.all;
use ieee.std_logic_arith.all;
use ieee.std_logic_unsigned.all;

--entity
entity logic_test is
port(
a,b,c,d   :in  std_logic;
out1,out2 :out std_logic
);
end entity logic_test;

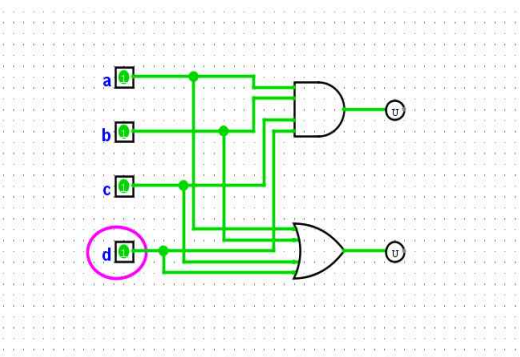
--architecture
architecture bhv of logic_test is
begin
out1 <= a and b and c and d;
out2 <= a or b or c or d;
end architecture bhv;
```

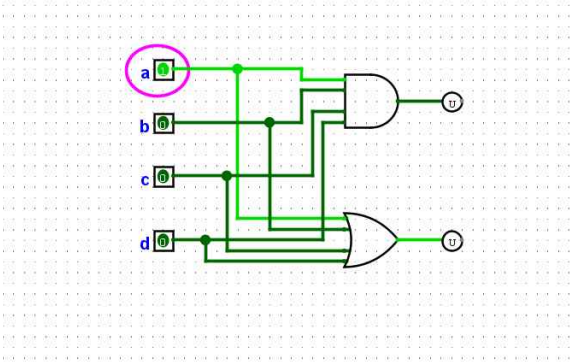
circuit

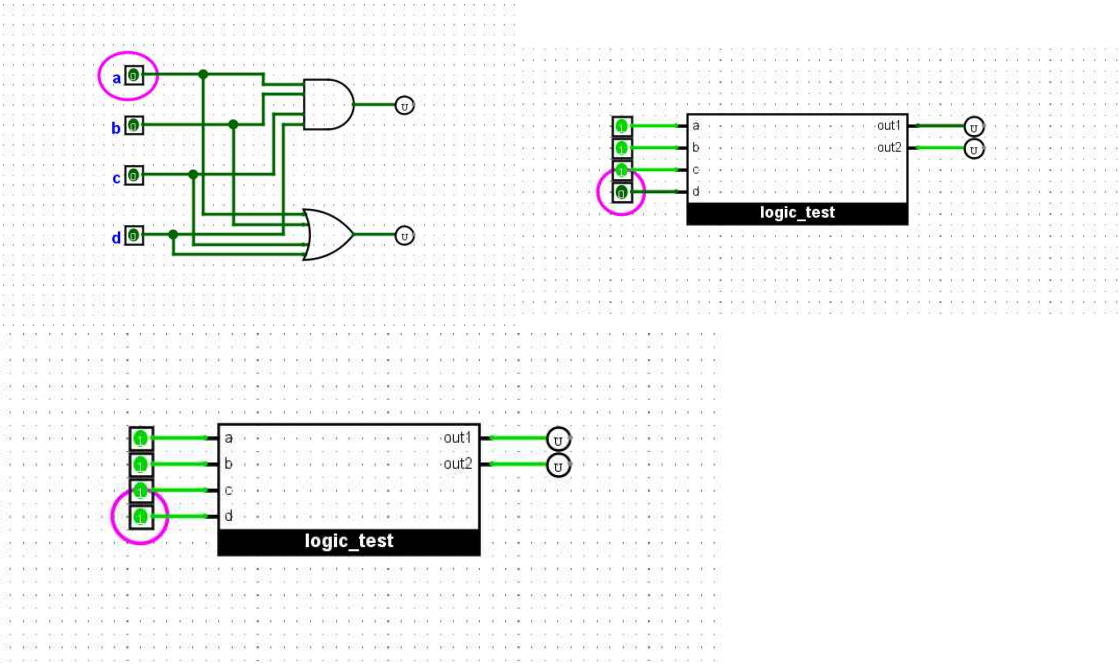




result







For inputs a, b, c, and d, the outputs are 'out1 <= a and b and c and d' and 'out2 <= a or b or c or d'. The above circuit outputs the result corresponding to the AND and OR truth tables.

2. 8-to-1 Multiplexer

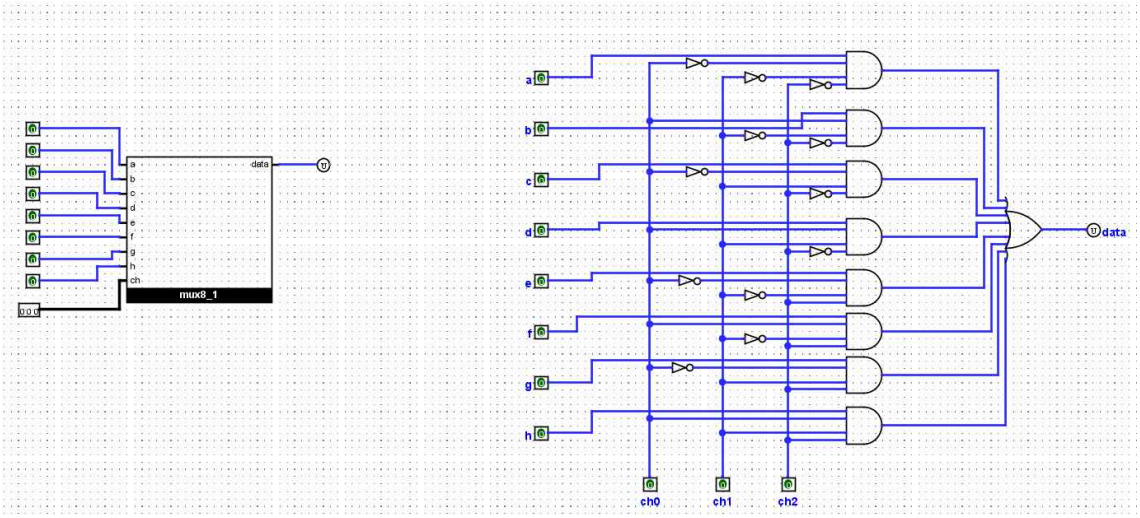
```
--library and package
library ieee;
use ieee.std_logic_1164.all;
use ieee.std_logic_arith.all;
use ieee.std_logic_unsigned.all;

--entity
entity mux8_1 is
port(
a,b,c,d,e,f,g,h :in  std_logic;
ch              :in  std_logic_vector(2 downto 0);
data            :out std_logic
);
end entity mux8_1;

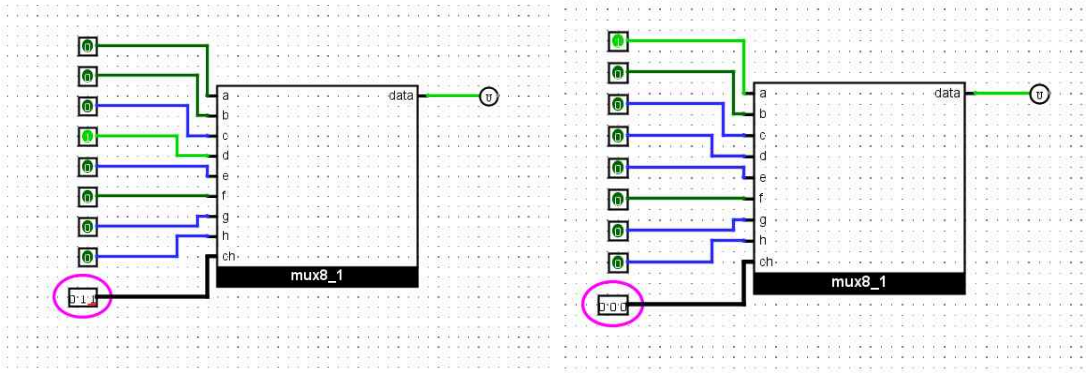
--architecture
architecture bhv of mux8_1 is
begin
process(a,b,c,d,e,f,g,h,ch)
begin
case ch is
when "000" => data <= a;
when "001" => data <= b;
when "010" => data <= c;
when "011" => data <= d;
when "100" => data <= e;
when "101" => data <= f;
when "110" => data <= g;
when "111" => data <= h;
```

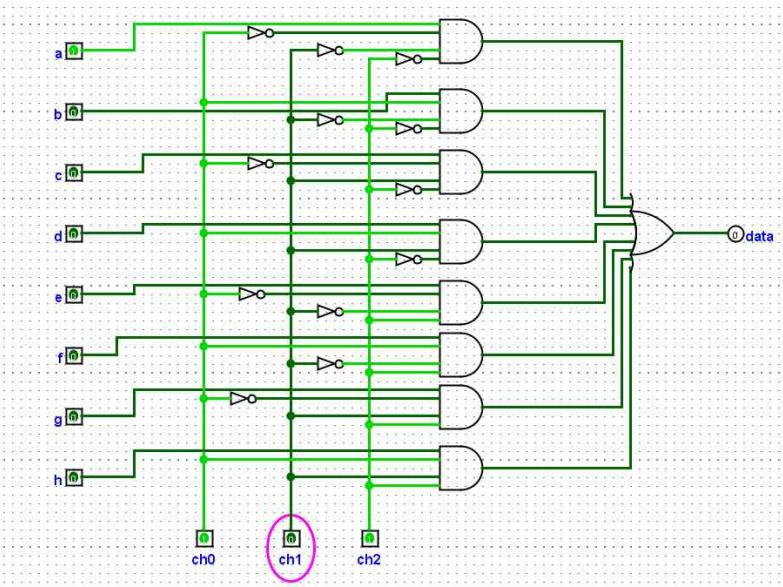
```
when others => data <= '0';
end case;
end process;
end architecture bhv;
```

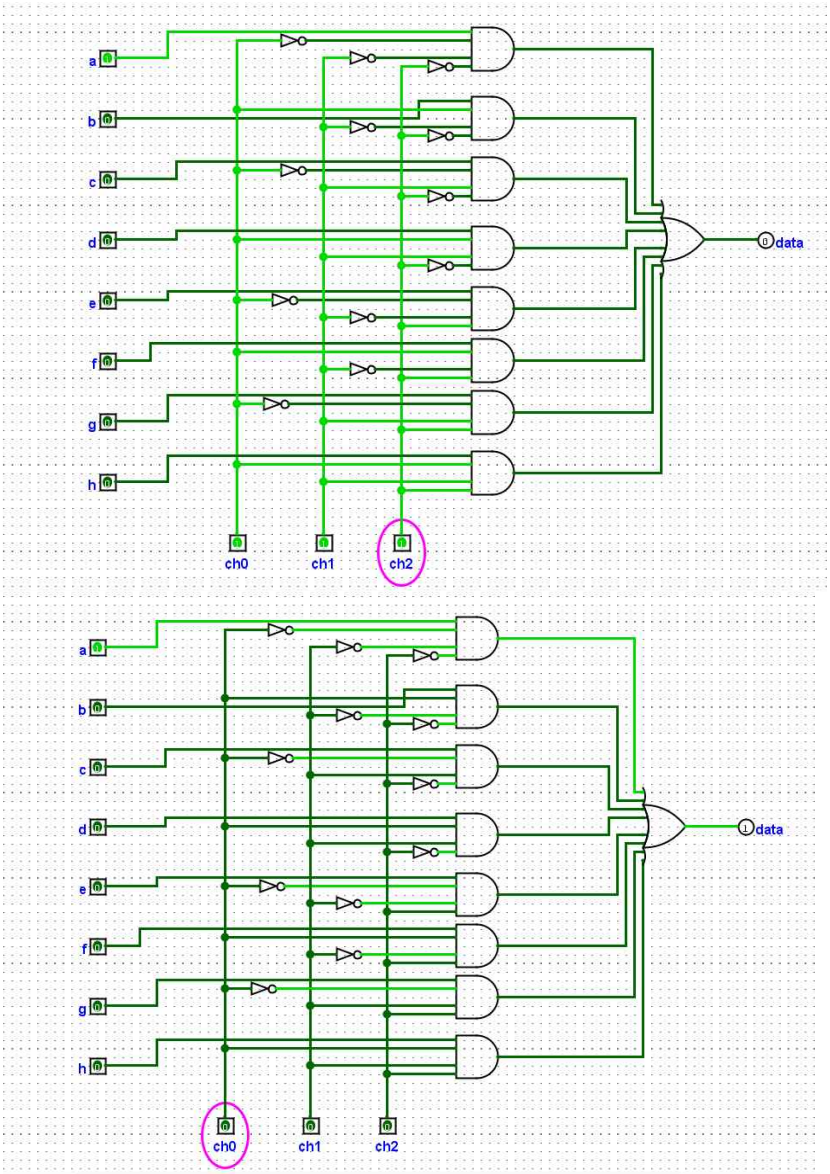
circuit



result







The circuit outputs one of the inputs a through h as data, according to the value of the 3-bit select signal ch. For example, data is d when ch is 011 and a when ch is 000. data is 1 only when the selected input is 1, and changing the unselected inputs does not affect the output, so the circuit works correctly as a multiplexer.

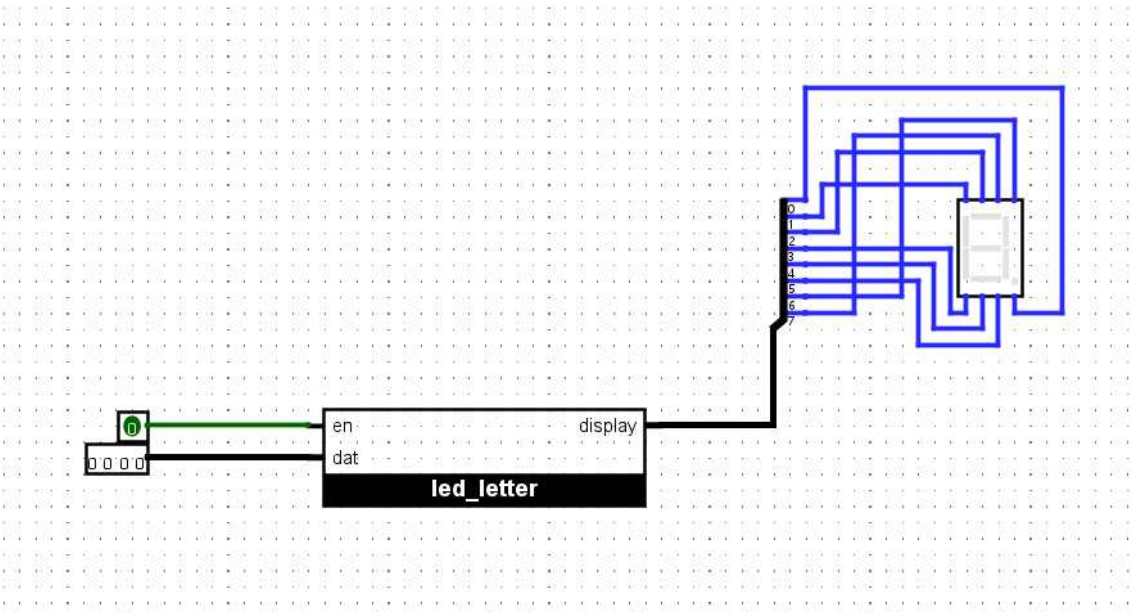
3. Use digital tubes to display letters: a,b,c,d,f,g

```
--library and package
library ieee;
use ieee.std_logic_1164.all;
use ieee.std_logic_arith.all;
use ieee.std_logic_unsigned.all;
```

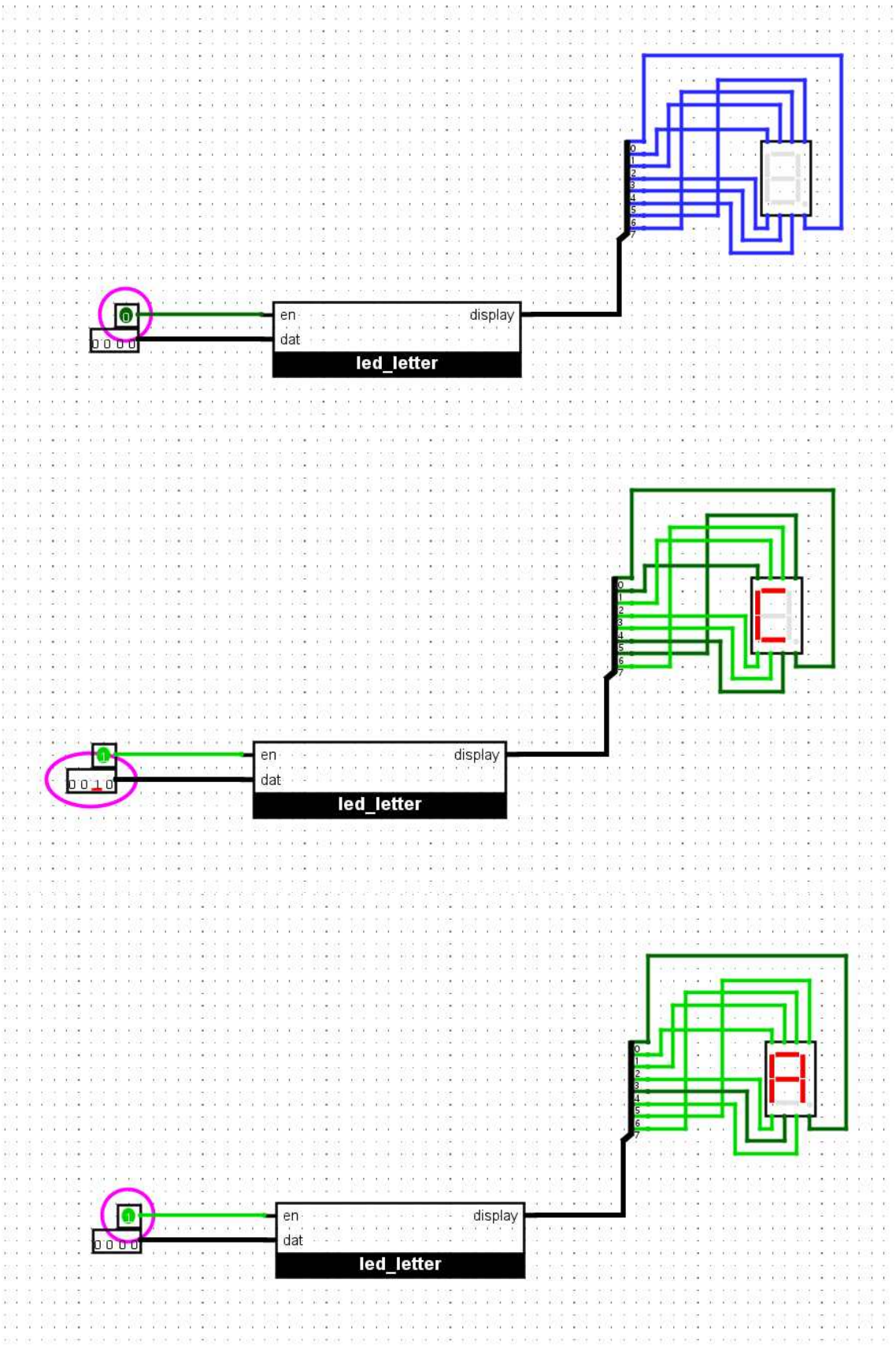
```
--entity
entity led_letter is
port(
en      :in  std_logic;
dat     :in  std_logic_vector(3 downto 0);
display :out std_logic_vector(7 downto 0)
);
end entity led_letter;

--architecture
--display = a b c d e f g dp
architecture bhv of led_letter is
begin
process(en,dat)
begin
if en='0' then
display <= "ZZZZZZZZ";
else
case dat is
when "0000" => display <= "11101110";  --a
               when "0001" => display <= "00111110";  --b
               when "0010" => display <= "10011100";  --c
               when "0011" => display <= "01111010";  --d
               when "0100" => display <= "10001110";  --f
               when "0101" => display <= "11110110";  --g
               when others => display <= "00000000";
end case;
end if;
end process;
end architecture bhv;
```

circuit



result


en
dat
display
led_letter
0 0 0 0
en
dat
display
led_letter
0 0 1 0
en
dat
display
led_letter
0 0 0 0

The led_letter entity converts the 4-bit input dat into a segment pattern for the 7-segment display. When en is 1 and dat is 0000 to 0101, the display shows A, b, C, d, F, and g, respectively. When en is 0, all segments are off. Since a 7-segment display cannot show every letter in the same case, A, C, and F are displayed in uppercase and b, d, and g in lowercase.

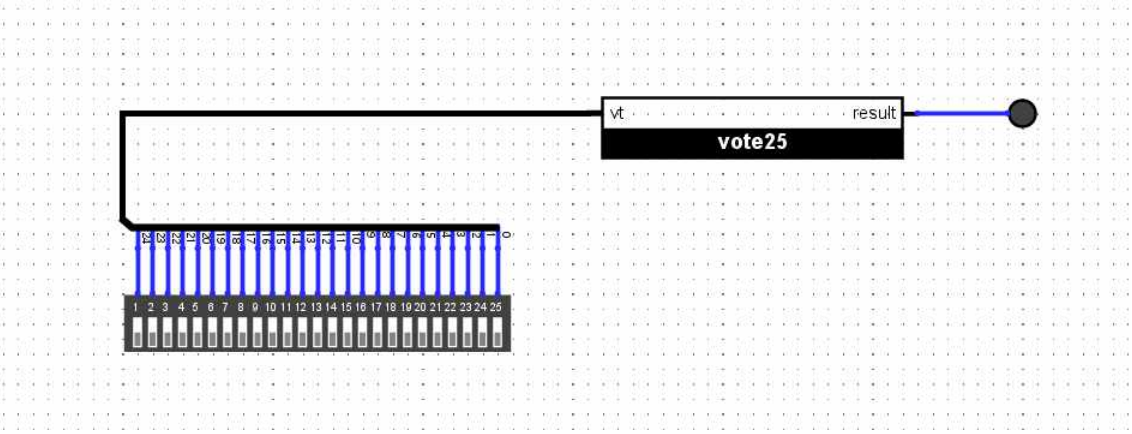
4. 25-person voting machine

```
--library and package
library ieee;
use ieee.std_logic_1164.all;
use ieee.std_logic_arith.all;
use ieee.std_logic_unsigned.all;

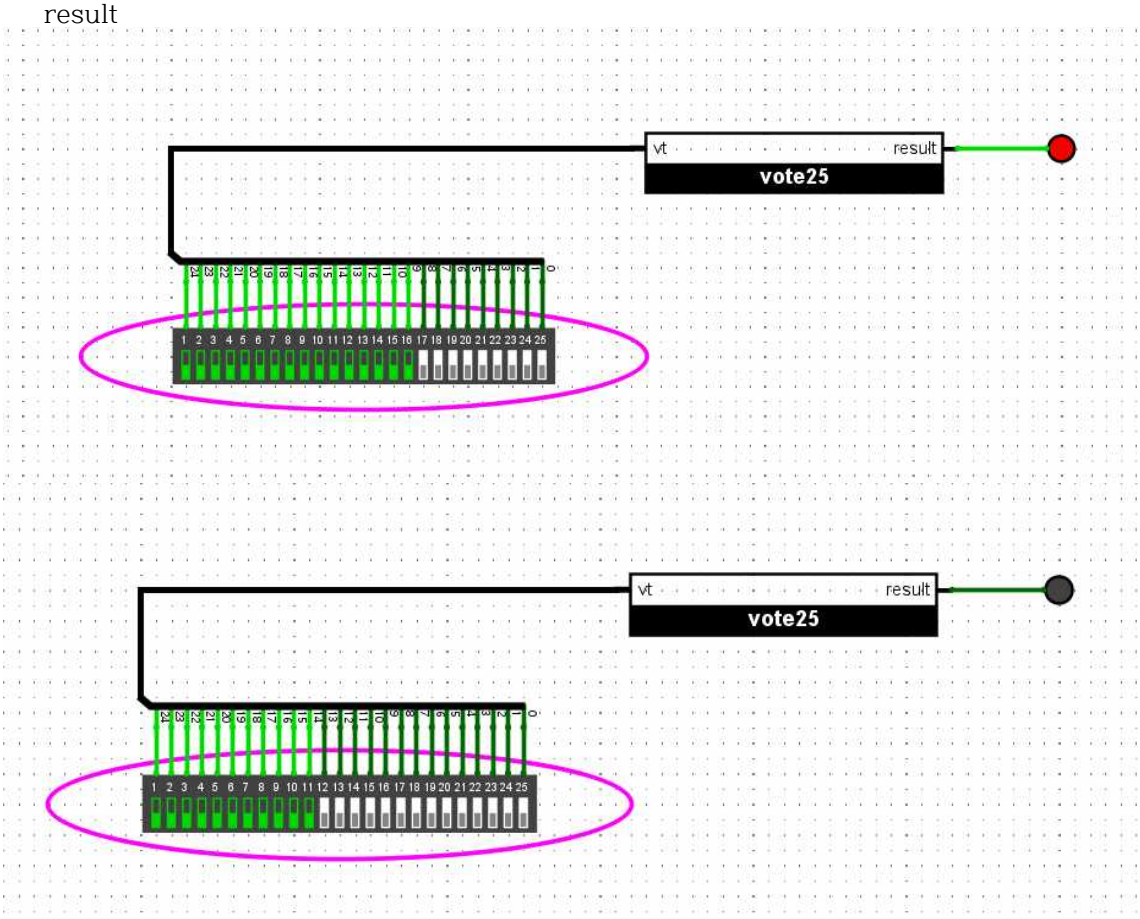
--entity
entity vote25 is
port(
vt     :in  std_logic_vector(25 downto 1);
result :out std_logic
);
end entity vote25;

--architecture
architecture bhv of vote25 is
begin
process(vt)
variable sum:integer range 0 to 25;
begin
sum:= 0;
for i in 1 to 25 loop
if (vt(i)='1') then
sum:=sum+1;
end if;
end loop;
if sum >= 13 then
result <= '1';
else
result <= '0';
end if;
end process;
end architecture bhv;
```

circuit



result



vote25 counts the number of 1s in the 25-bit input vt and outputs result = 1 if the count is 13 or more, and 0 otherwise. result was 0 when 11 switches were on and 1 when 16 switches were on, so the circuit works as expected for the majority condition (13 or more).

5. Design a VHDL module that counts the number of '1's in an input signal 'vt'(a 13-bit vector) and displays the corresponding number or letter on a 7-segment display.

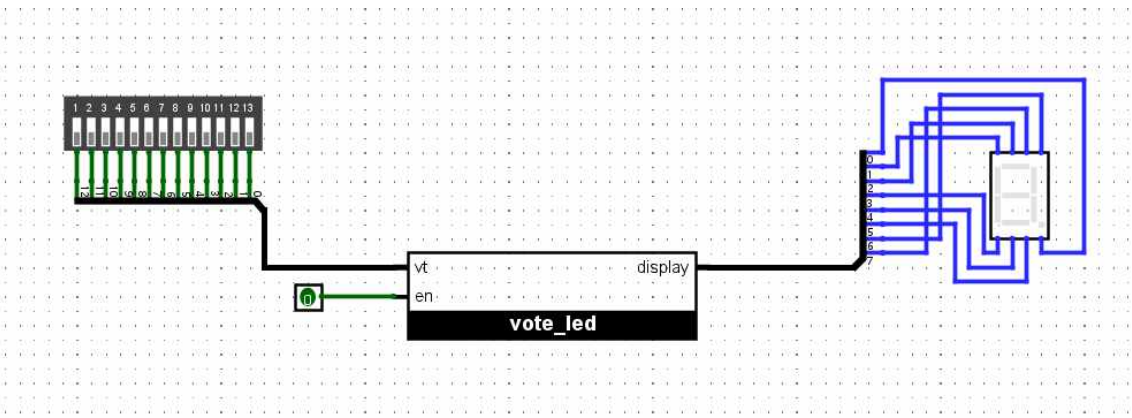
```
--library and package
library ieee;
use ieee.std_logic_1164.all;
use ieee.std_logic_arith.all;
use ieee.std_logic_unsigned.all;

--entity
entity vote_led is
port(
vt      :in  std_logic_vector(13 downto 1);
en      :in  std_logic;
display :out std_logic_vector(7 downto 0)
);
```

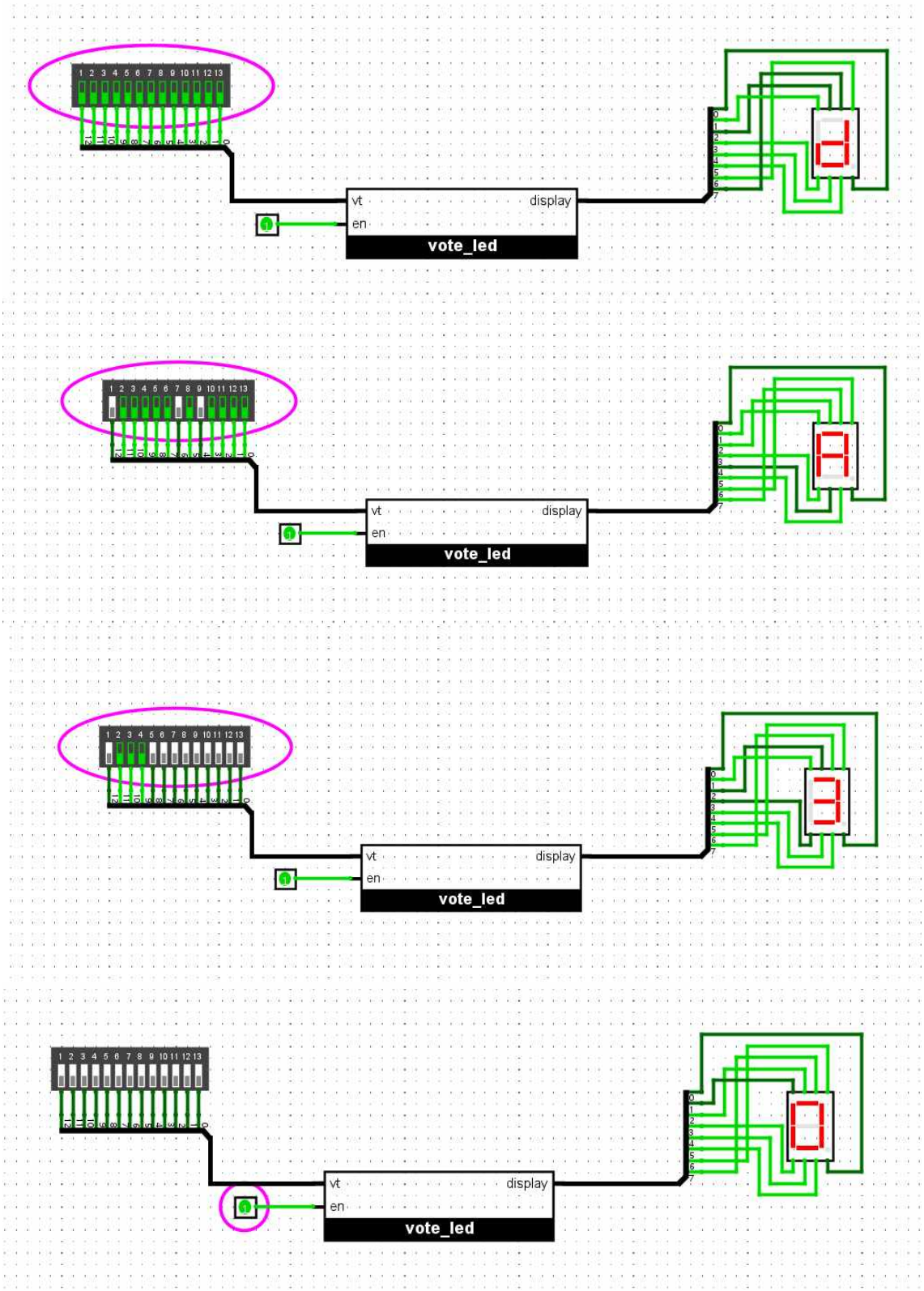
```
end entity vote_led;

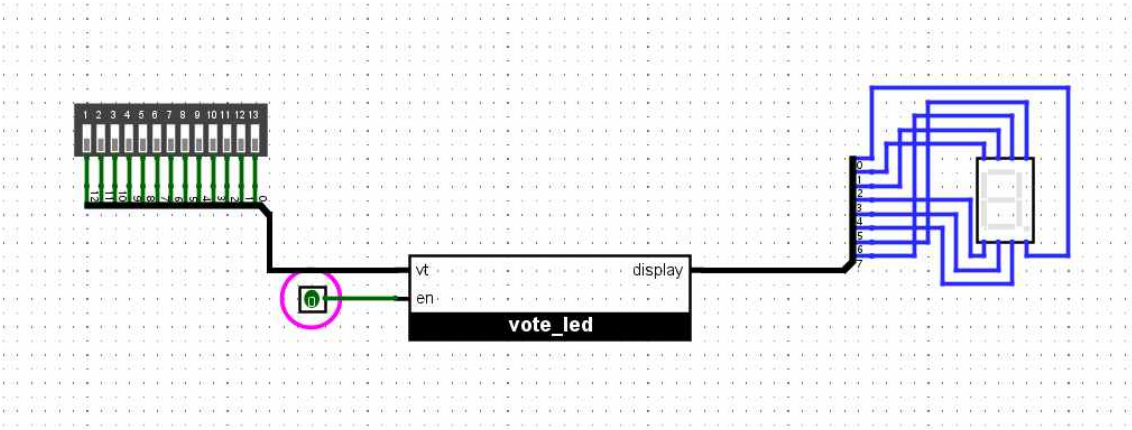
--architecture
--display = a b c d e f g dp
architecture bhv of vote_led is
begin
process(vt,en)
variable sum:integer range 0 to 15;
begin
sum:= 0;
for i in 1 to 13 loop
if (vt(i)='1') then
sum:=sum+1;
end if;
end loop;
if en='0' then
display <= "ZZZZZZZZ";
else
case sum is
when 0  => display <= "11111100";  --0
              when 1  => display <= "01100000";  --1
              when 2  => display <= "11011010";  --2
              when 3  => display <= "11110010";  --3
              when 4  => display <= "01100110";  --4
              when 5  => display <= "10110110";  --5
              when 6  => display <= "10111110";  --6
              when 7  => display <= "11100000";  --7
              when 8  => display <= "11111110";  --8
              when 9  => display <= "11100110";  --9
              when 10 => display <= "11101110";  --A
              when 11 => display <= "00111110";  --B
              when 12 => display <= "10011100";  --C
              when 13 => display <= "01111010";  --D
              when 14 => display <= "10011110";  --E
              when 15 => display <= "10001110";  --F
              when others => display <= "00000000";
end case;
end if;
end process;
end architecture bhv;
```

circuit



result


vt
en
display
vote_led
vt
en
display
vote_led
vt
en
display
vote_led
vt
en
display
vote_led



The circuit counts the number of 1s in the 13-bit input vt and displays 0-9 as digits and 10-13 as A, b, C, and d on a 7-segment display. When en is 0, the display is off, and when en is 1 the result is shown. With 13, 10, 3, and 0 switches on, the display showed d, A, 3, and 0, respectively. Since vt has only 13 bits, the count never exceeds 13, so E (14) and F (15), which the problem requires, are included in the code but can never be displayed.